



## GOVERNMENT OF INDIA

### OFFICE OF THE DIRECTOR GENERAL OF CIVIL AVIATION

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**Subject: Guidance material for training and endorsement of VTOL capable aircraft (VCA) rating.**

#### 1. Introduction

- 1.1 Advanced Air Mobility (AAM) is an emerging aviation ecosystem that leverages new aircraft and an array of innovative technologies that would be able to provide more efficient, more sustainable, and more equitable mode of transportation in future. These new aircrafts are having new aircraft design and system technologies. While some of the new designs may resemble traditional winged aircraft, some are anticipated to include powered lift and Vertical Take-off and Landing (VTOL) capabilities that facilitate operations between desired locations.
- 1.2 This guidance material is aimed to provide minimum training and endorsement requirement for VTOL capable aircraft (VCA) rating and requirements for appointment/ authorization of examiner & instructor for VCA.
- 1.3 This guidance material is issued under provisions of Rule 38, 38A & 41A of the Aircraft Rules, 1937.
- 1.4 The requirements laid down in this circular will be reviewed after a period not exceeding three years from the date of publication of this circular.

#### 2. Definition

***Vertical take-off and landing (VTOL)-capable aircraft (VCA)*** means a power-driven, heavier than-air aircraft, other than aeroplane or rotorcraft, capable of performing vertical take-off and landing by means of lift and thrust units used to provide lift during take-off and landing.

#### 3. Acronyms

|      |   |                                                                   |
|------|---|-------------------------------------------------------------------|
| AAM  | - | Advanced Air Mobility                                             |
| AFM  | - | aircraft flight manual                                            |
| ATO  | - | Approved Training Origination                                     |
| FSB  | - | Flight Safety Board Report issued by FAA                          |
| FSTD | - | Flight Synthetic Training Device                                  |
| ICAO | - | International Civil Aviation Organization                         |
| OEB  | - | Operation evaluation Board Report issued by Transport Canada/ANAC |
| OEM  | - | Original Equipment Manufacture                                    |

|           |   |                                           |
|-----------|---|-------------------------------------------|
| OSD       | - | Operation Suitability Data issued by EASA |
| SARPS     | - | standards and recommended practices       |
| SCA       | - | STOL capable aircraft                     |
| STOL      | - | Short Take-Off and Landing                |
| VCA       | - | VTOL capable aircraft                     |
| TRE (VCA) | - | Type rated Examiner Instructor (VCA)      |
| TRI (VCA) | - | Type rated Instructor (VCA)               |
| VFR       | - | Visual flight rules                       |
| VMC       | - | Visual meteorological conditions          |

#### **4. General Requirements**

- 4.1** No person shall fly as a pilot of a VTOL-capable aircraft which is not included or entered in the aircraft rating of the licence except, as permitted under Rules 6B and 6C of the Aircraft Rules, 1937.
- 4.2** The licence shall indicate the types of VTOL-capable aircraft the holder is entitled to fly. In order to endorse an additional type of VTOL-capable aircraft, an applicant need to undergo a training course, flight instruction followed by skill test in accordance with the syllabus defined by the OEM or approved by DGCA.
- 4.3** The VTOL-capable aircraft type rating training course shall include the mandatory training elements for the relevant type as defined in the OSD/OEB/FSB.
- 4.4** Initial type Rating Training shall be carried out under scope of approval of OEM or ATO approved by DGCA/ICAO contracting State.
- 4.5** Ground and simulator training/ flight instruction shall be completed in the same ATO.
- 4.6** The skill test of the trainee pilot shall be carried out by an examiner TRE (VCA), other than those who had imparted the training. Training and checks must be carried out by TRI (VCA)/ TRE (VCA) approved by DGCA/ concerned contracting state/ OEM.

#### **5. Prerequisites for VTOL-capable aircraft (VCA) Type Rating**

- 5.1** An applicant for a VTOL-capable aircraft rating shall comply with the following experience requirements and prerequisites, before starting VCA Type Rating training, unless otherwise determined in the OSD/OEB/FSB:
  - a) Hold a current (as defined in para 5.2 below) and valid CPL (A/H)/ATPL (A/H) issued under the Aircraft Rules, 1937; and
  - b) VALID CLASS –I Medical assessment issued under the Aircraft Rules, 1937.
  - c) Prior Minimum Experience in turbo prop or turbo Jet if determined by OEM or vertical lift/ helicopter (if determined by OEM).
- 5.2** For the purpose of this circular “current” is defined as:
  - a) Having at least 10 hours as flying experience on Indian Registered Aeroplane/ helicopter endorsed in the CPL (A/H)/ATPL (A/H) of at least 10 hours as Pilot in Command in the preceding twenty four months from the date of application. (Fifty percent of Co Pilot maybe counted towards requirement of 10 hours PIC);

Or

- b) Pilot Proficiency Check/ Skill Tests/ Flight Review on the Aeroplane/ helicopter endorsed in the CPL (A/H)/ATPL (A/H) in the preceding twenty four months from the date of application for the VTOL-capable aircraft (VCA) Type Rating.

## **6. Requirements for issue of VTOL-capable aircraft Type Rating**

- 6.1 Training course:** An applicant for a VTOL-capable aircraft (VCA) type rating shall complete a training course and flight instruction at an ATO approved by the DGCA or any ICAO contracting state/ OEM.
- 6.2 Medical Standards:** The medical of the trainee pilot shall be valid during the training and checks unless training is done on synthetic training device representing VTOL-capable aircraft.
- 6.3 Theoretical knowledge:** The training course for VTOL-capable aircraft (VCA) Type ratings shall be as approved by DGCA/ concerned contracting state/ OEM.
- 6.4 Theoretical knowledge examination:** The applicant for issue of a VTOL-capable aircraft (VCA) type rating shall pass a theoretical knowledge examination organized by the ATO to demonstrate the level of theoretical knowledge required for the safe operation of the applicable VTOL-capable aircraft (VCA) type. The trainee pilot should have passed technical examination on type before commencement of the flying Instruction/training.
- 6.5 Flight instruction for VTOL-capable aircraft (VCA) Type ratings:** The training course for flight instruction in VTOL-capable aircraft (VCA) Type ratings shall be as approved by DGCA/ concerned contracting state/ OEM.
- 6.6 Skill Test:** An applicant for a VTOL-capable aircraft type rating shall pass a skill test in accordance with manner prescribed by Director General to demonstrate the skill required for the safe operation of the applicable VTOL-capable aircraft (VCA).
- 6.7** The applicant shall pass the skill test within a period of 6 months after commencement of the type rating training course.

## **7. Privilege and limitation of holder of a VTOL-capable aircraft type rating**

- a) The privileges of the holder of a VTOL-capable aircraft type rating is to act as pilot, on type of VTOL-capable aircraft specified in the aircraft rating of his/her licence.
- b) VTOL-capable aircraft type rating shall only be exercised during the day VFR condition.

## **8. Privilege on variant of a VTOL-capable aircraft type**

- a) Pilots shall complete differences training/ familiarisation (as prescribed by OSD/EEB/FSB) in order to extend their privileges to another variant of aircraft within one VTOL-capable aircraft type rating.
- b) The differences training shall be conducted at any of the following:
  - i. An ATO approved by DGCA/ICAO contracting state/ OEM; or
  - ii. An Indian AOC holder having an approved differences training programme for the relevant class or type.
- c) If pilots have not flown the variant within 2 years following the training listed above in para 9 b), a further differences training or a proficiency check in that variant shall be completed.
- d) The differences training or the proficiency check on variant shall be entered in the pilots' logbook or equivalent record and signed by the examiner/ instructor.

## **9. Designation of PIC**

No VCA operator may use a person, nor may any person serve, as a pilot in command of a VTOL-capable aircraft unless he meets the following requirements laid down in approved OM:

- a) Minimum flight crew experience requirements.
- b) Operating experience required on type VTOL-capable aircraft.
- c) Demonstrate to the operator an adequate knowledge, skill of the route and vertiports to be flown.

## **10. Recency requirement of a VTOL-capable aircraft**

- (a) A pilot shall not operate a VTOL-capable aircraft in commercial air transport or to carry passengers as PIC or co-pilot unless he or she has carried out, in the preceding 90 days, at least 3 take-offs, approaches and landings as a pilot flying (PF) in a VTOL-capable aircraft of the same type or synthetic training device representing that type of a VTOL-capable aircraft. take-offs and landings
- (b) In the case of a gap in flying, applicants must undergo following training to regain the proficiency to safely operate the VTOL-capable aircraft of the specific type before exercising their privileges. This training must be completed either at an ATO/OEM or with an Indian AOC holder operator that is specifically approved for such training;

| Break in Flying                    | Technical and Performance Refresher | Flying training with TRE/TRI on VCA <u>OR</u> on Simulator approved by DGCA/ ICAO Contracting State/ OEM                                                  |
|------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 90 days to less than 6 months year | -                                   | 3 take-offs and landings, without passengers on board in 00:45 Hrs (When in Simulator same may be achieved in 01 session, Total 00:45 hr.)                |
| 6 months to 1 year                 | 1:00 hrs                            | 3 take-offs and landings without passengers on board in 00:45 Hrs. and PPC if due (When in Simulator same may be achieved in 01 session, Total 00:45 hr.) |
| 1 year to less than 2 years        | 8:00 hrs                            | Dual 00:45 hr. and Skill Test (3 take-offs and landings) 00:45 hr. (When in Simulator same may be achieved in 02 sessions, Total 1:30 hr.)                |
| 2 years to less than 3 years       | 12:00 hrs                           | Dual 1:30 hr and Skill Test (3TOL) 00:45 hr. (when in Simulator same may be achieved in 03 sessions, Total 2:15 hr)                                       |

- (c) The provisions of sub para (b) shall not be applicable in case a pilot has not exercised the privileges of his license for a period of 3 years or more on VCA, in such a case the pilot shall undergo the training required for initial type rating.

#### 11. Currency requirement of a VTOL-capable aircraft (VCA)

The holder of VTOL-capable aircraft rating is required to comply with the requirements of Rules 39B (medical standards), recent flight experience as per para 10 above and valid PPC before exercising the privileges on that type of VTOL-capable aircraft.

#### 12. Proficiency Check (PPC)

- In order to act as a PIC/Co-pilot with VCA Operator, the VTOL-capable aircraft rating holder shall be required to undergo proficiency check as specified by the Director General, in respect of the type of VTOL-capable aircraft (VCA) to be flown.
- The proficiency check carried out as per para (a) shall be valid for a period of six months from the date of the check.
- In the case of a renewal, the period of validity shall commence from the date of expiry of the previous validity provided that the check has been carried out within two months preceding the date of expiry.

#### 13. Requirements for issuance of authorization of TRI (VCA)

An applicant for a **TRI (VCA)** shall comply with the following experience requirements and prerequisites, unless otherwise determined in the OSD/OEB/FSB:

- a) Hold a current and valid *TRI (A/H)* issued under the Aircraft Rules, 1937; and
- b) Hold a current and valid a VTOL-capable aircraft rating on the same type on which TRI is requested.
- c) Be entitled to act as PIC in VTOL-capable aircraft rating during flight instruction.
- d) *A TRI is considered to be current only if instructor have during last one year received instructor refresher training at an ATO during which the competency required to instructor on a course is assessed to the satisfaction of the Examiner.*
- e) Have completed Instructor training course at ATO/ OEM/ VCA operator, and theoretical and practical training shall including mandatory training elements as specified in the operation suitability data or flight Safety Board report.
- f) Pass the instructor assessment of competency by the TRE (VCA).
- g) Flying Experience:
  - i. Total (A/H) experience - 1000 hours
  - ii. PIC on Type (VCA) - 300 hours
  - iii. Within preceding 12 months, minimum 30 hours of the route sector as PIC on Type VTOL-capable aircraft (VCA) or 15 hour route sector maybe completed in in a FSTD representing that VCA type.

#### **14. Privileges of the TRI (VCA) authorization**

No pilot shall exercise the privilege of a TRI (VCA) unless he / she is duly approved by DGCA and meets all the applicable requirements of this circular. The privileges of the TRI (VCA) are as following:

- a) Training for VTOL-capable aircraft (VCA) type rating;
- b) Base training;
- c) Conduct of PPC
- d) Initial Operating experience required on type VTOL-capable aircraft (VCA)

#### **15. Recent Experience Requirements for TRI (VCA)**

##### **15.1 TRI shall exercise his/ her privileges only when:**

- a) Have undergone refresher training and assessment to the satisfaction of TRE (VCA).
- b) Has a current and valid assessment of TRI competency and meets the recent experience requirement of five (05) hours PIC in the last 90 days on the same type of VCA.
- c) Has exercised the privilege of TRI, on the same type within the preceding 12 month, except when a pilot is newly approved on a type of VCA.

##### **15.2 In case the TRI (VCA) is not meeting the aforesaid recent experience requirements of para 15.1 above, he/ she shall first exercise his functions under the supervision of a TRE (VCA). He/ she shall start functioning independently again, only if**

proficiency is found satisfactory and the same has been certified by the TRE (VCA) in the logbook.

- 15.3** A certificate indicating compliance of the aforesaid recent experience requirements shall be recorded by all TREs in the Test / Check / Training reports.

**16. Validity of the TRI (VCA) authorization**

Unless revoked or suspended, the approval of the TRI (VCA) shall remain valid for a period of three (03) years from the date of issue of approval so long as the pilot remains in the employment of the operator who has requested and obtained the approval.

**17. Renewal of TRI on VTOL-capable aircraft (VCA)**

Applicants shall, within the 12 months immediately preceding the date of the application, have passed the assessment of competence in accordance with this circular and shall have completed:

- a) At least 10 hours of flight time, including take-offs and landings on the applicable type, of which maximum 5 hours may be completed on synthetic training device prescribed by OEM/FSB/OSD.
- b) Completed Instructor refresher training that focus on the privilege of that VCA.
- c) Pass the relevant section of the assessment of competency.

**18. Special conditions**

- 18.1** The Director General may issue a specific certificate granting privilege for flight instruction when compliance with the requirements established in this circular is not possible in case of introduction of new type of VTOL-capable aircraft (VCA).

- 18.2** Such a certificate shall be restricted to the training flights required for the introduction of the new type of VTOL-capable aircraft (VCA), provided the individual meets the following criteria:

- a) Total flying experience(A/H) - 2000 hours
- b) TRI(A/H) - 500 hours
- c) PIC on Type (VCA) - 100 hours
- d) Completion of the instructor course at the DGCA/ ICAO contracting State approved ATO/ OEM.

- 18.3** The validity of this authorization shall not, in any case, exceed one year.

**19. Requirements for the appointment of TRE (VCA)**

An applicant for a **TRE (VCA)** shall comply with the following experience requirements and prerequisites, unless otherwise determined in the OSD/OEB/FSB:

- a) Hold a current and valid *TRE (A/H)* issued under the Aircraft Rules, 1937; and
- b) Hold a current and valid a VTOL-capable aircraft rating on the same type on which TRE is requested.
- c) Be entitled to act as PIC in VTOL-capable aircraft rating during flight instruction.
- d) *A TRE is considered to be current only if instructor have during last one year received instructor refresher training at an ATO during which the competency required to instructor on a course is assessed to the satisfaction of the Examiner.*
- e) Have completed Examiner standardisation course at ATO/ OEM/ VCA operator, and theoretical and practical training shall including mandatory training elements as specified in the operation suitability data or flight Safety Board report.
- f) Pass the examiner assessment of competency by the TRE (VCA)/ FOI.
- g) Flying Experience:
  - i. Total flying experience (A/H) - 1500 hours
  - ii. PIC on Type (VCA) - 300 hours
  - iii. Within preceding 12 months, minimum 30 hours of the route sector as PIC on Type VTOL-capable aircraft (VCA) or 15 hour route sector maybe completed in in a FSTD representing that VCA type.

## **20. Privilege of the TRE (VCA)**

No pilot shall exercise the privilege of a TRE (VCA) unless he / she is duly approved by DGCA and meets the requirements of this circular. The privileges of the TRE (VCA) on shall be as follows for VTOL-capable aircraft (VCA):

- a) Skill tests for (VCA) type rating
- b) All privileges of a TRI (VCA)
- c) Instruct the TRI (VCA)
- d) Assessment of competency for TRI (VCA)

## **21. Recent Experience Requirements for TRE (VCA)**

A TRE shall exercise his privilege only when:

- a) He has a current and valid assessment of TRE competency and meets the recent experience requirement of five (05) hours PIC in the last 90 days on the same type.
- b) He has exercised the privilege of TRE, on the same type within the preceding 12 month and at least conducted 5 skill test / PPC acts, except when a pilot is newly approved on a type of VCA.
- c) A certificate indicating compliance of the aforesaid recent experience requirements shall be recorded by all TREs in the Test / Check / Training reports.
- d)



## **22. Validity of the TRE (VCA) authorization**

Unless revoked or suspended, the approval of the TRE shall remain valid for a period of three (03) years from the date of issue of approval so long as the pilot:

- a) is current on the type of VTOL-capable aircraft (VCA) on which the approval has been granted;
- b) meets the applicable requirements of this circular;
- c) Remains in the employment of the operator who has requested and obtained the approval.

## **23. Renewal of TRE on VTOL-capable aircraft (VCA)**

Applicants shall, within the 12 months immediately preceding the date of the application, have passed the assessment of competence in accordance with this circular and shall have completed:

- a) at least 10 hours of flight time, including take-offs and landings on the applicable aircraft type, of which maximum 5 hours may be completed on synthetic training device.
- b) Completed an examiner refresher course training that focus on the privilege of that on VTOL-capable aircraft (VCA).
- c) Pass the relevant section of the assessment of competency.

## **24. Special condition**

**24.1** The Director General may issue a specific certificate granting privilege for TRE (VCA) when compliance with the requirements established in this circular is not possible in case of introduction of new type of VTOL-capable aircraft (VCA).

**24.2** Such a certificate shall be restricted to the privileges of TRE (VCA) required for the introduction of the new type of VTOL-capable aircraft (VCA), provided the individual meets the following criteria:

- a) Total flying experience (A/H) - 2500 hours
- b) TRE (A/H) experience - 500 hours
- c) PIC on Type (VCA) - 100 hours
- d) Completion of the examiner standardisation course at the DGCA/ ICAO contracting State approved ATO/ OEM.

**24.3** The validity of this authorization shall not, in any case, exceed one year.

## **25. Withdrawal of TRE (VCA) / TRI (VCA) Privileges**

**25.1** Approval granted to a pilot as TRE (VCA) / TRI (VCA) may be suspended or Cancelled/ withdrawn by DGCA if the pilot is found lacking in any of the requirements or if found unfit in any manner or in case of serious safety violation. The Accountable Manager/ Head of Training of the operator may also recommend to the DGCA, withdrawal of privileges as TRE (VCA) / TRI (VCA) giving adequate justification.

- 25.2** Except where there is an immediate threat to safety, the DGCA, prior to making a final decision in the matter of withdrawal of a TRE (VCA) / TRI (VCA) privileges, shall ensure the matter has been investigated thoroughly; and the TRE (VCA) / TRI (VCA) and, where applicable, the concerned Operator, have been given a formal opportunity to respond to the allegations, either verbally or in writing.

Sd/-

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